Advanced Formal Tools

CTL Modle Checker

Jing Wen

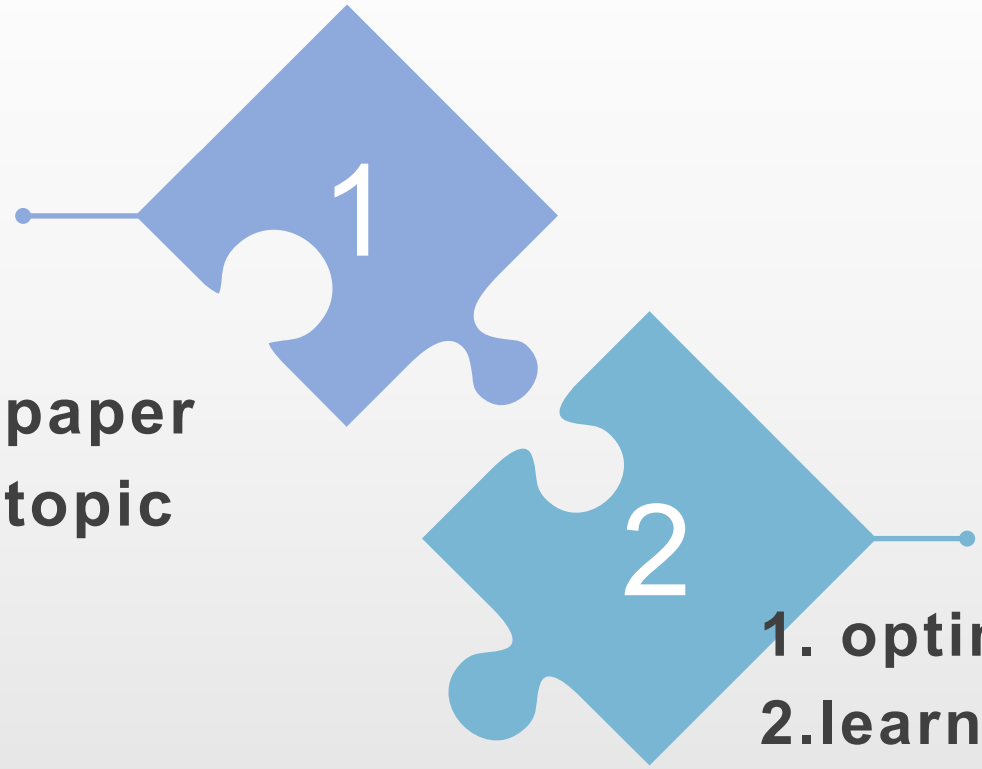
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CONTENTS

Current Status

1. Understanding of paper
2. Understanding of topic
3. What we did



Goal

1. optimize what we did
2. learn advanced coding strategies
3. Implement our own case study

CTL model checking algorithm

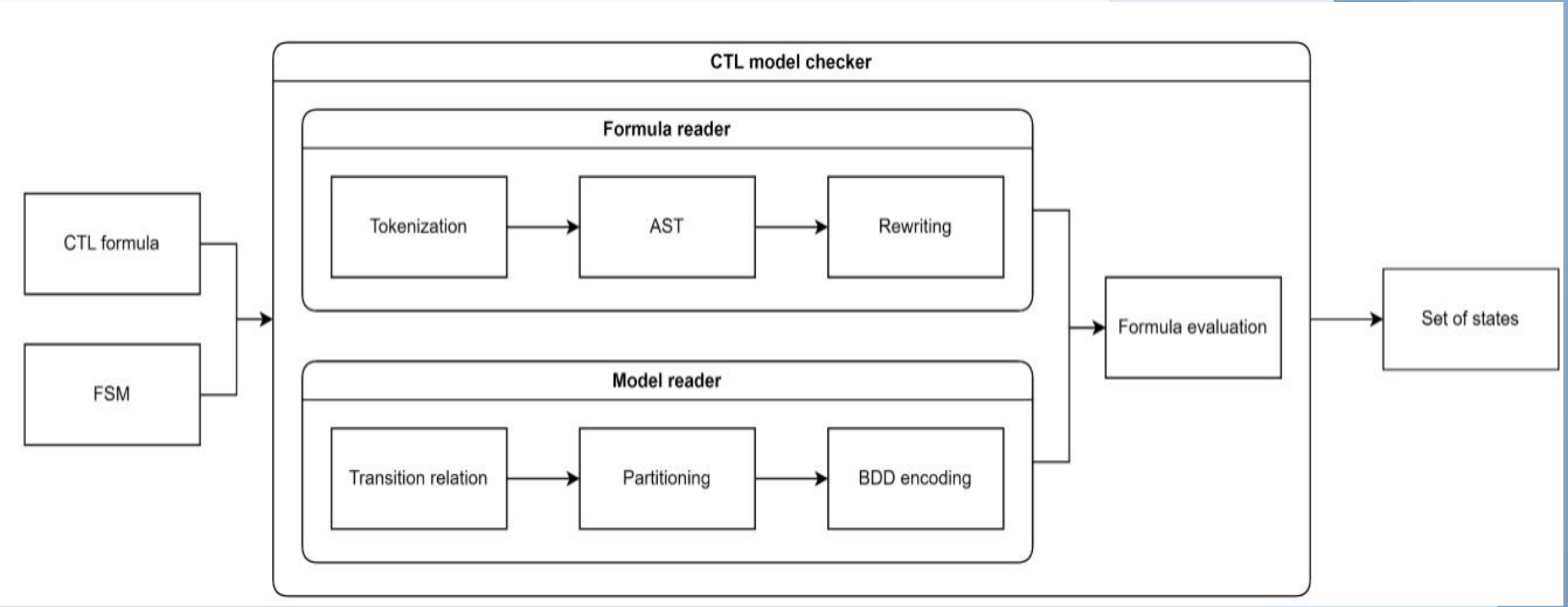
--based on forward state traversal

method:
The author presents a CTL model checking algorithm based on forward state traversal,
and combine it with BDD-based state traversal techniques by using partitioned transition relations.

Value: verify large FSMs with smaller time

CTL model checking algorithm

--based on forward state traversal



What we did

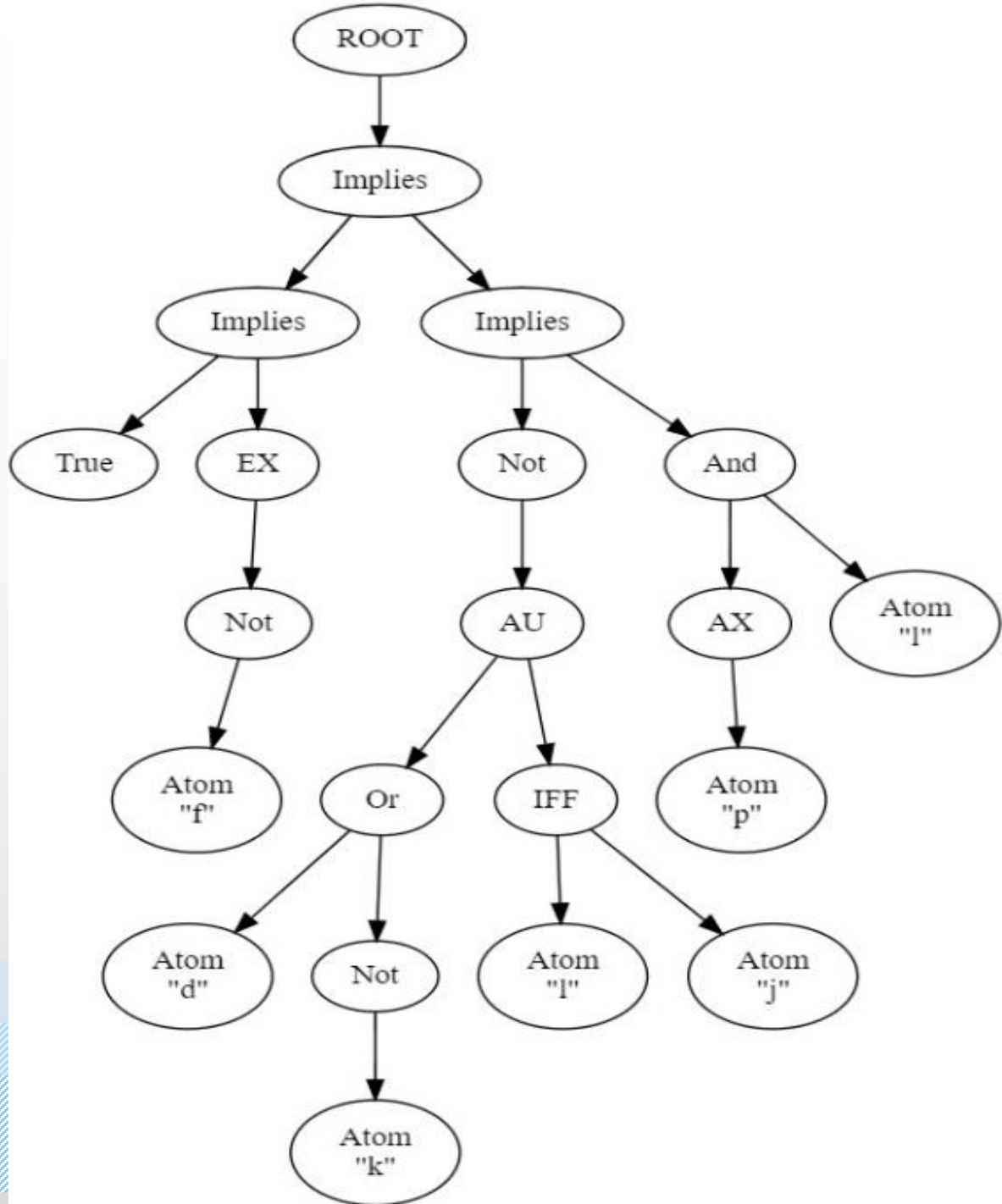
1. Program CTL parse to AST

example:

((true⇒EX(¬f))

⇒(¬A[(d∨(¬k))U(l⇔j)])

⇒(AXp∧l)



What we did

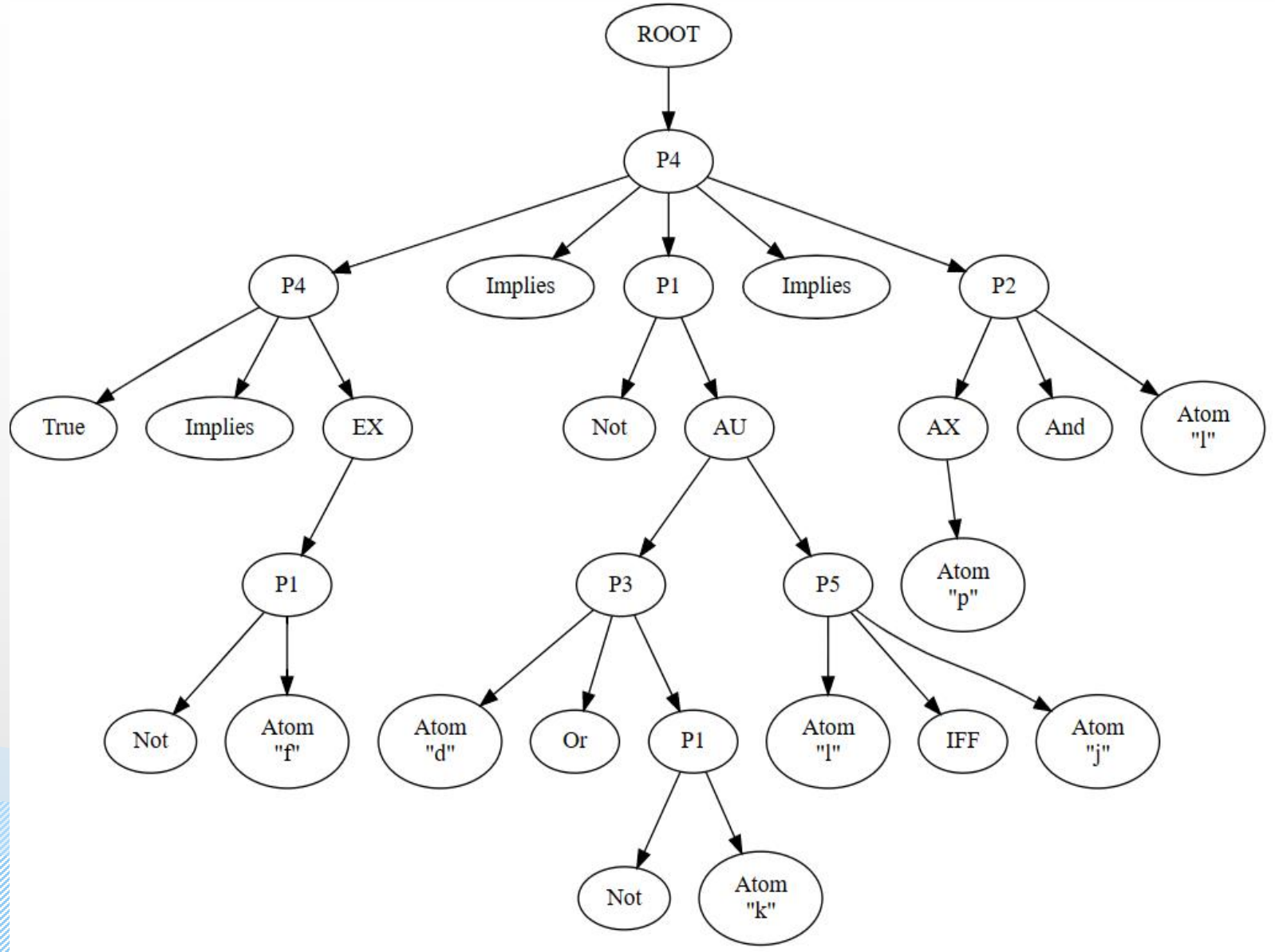
2. Optimize AST

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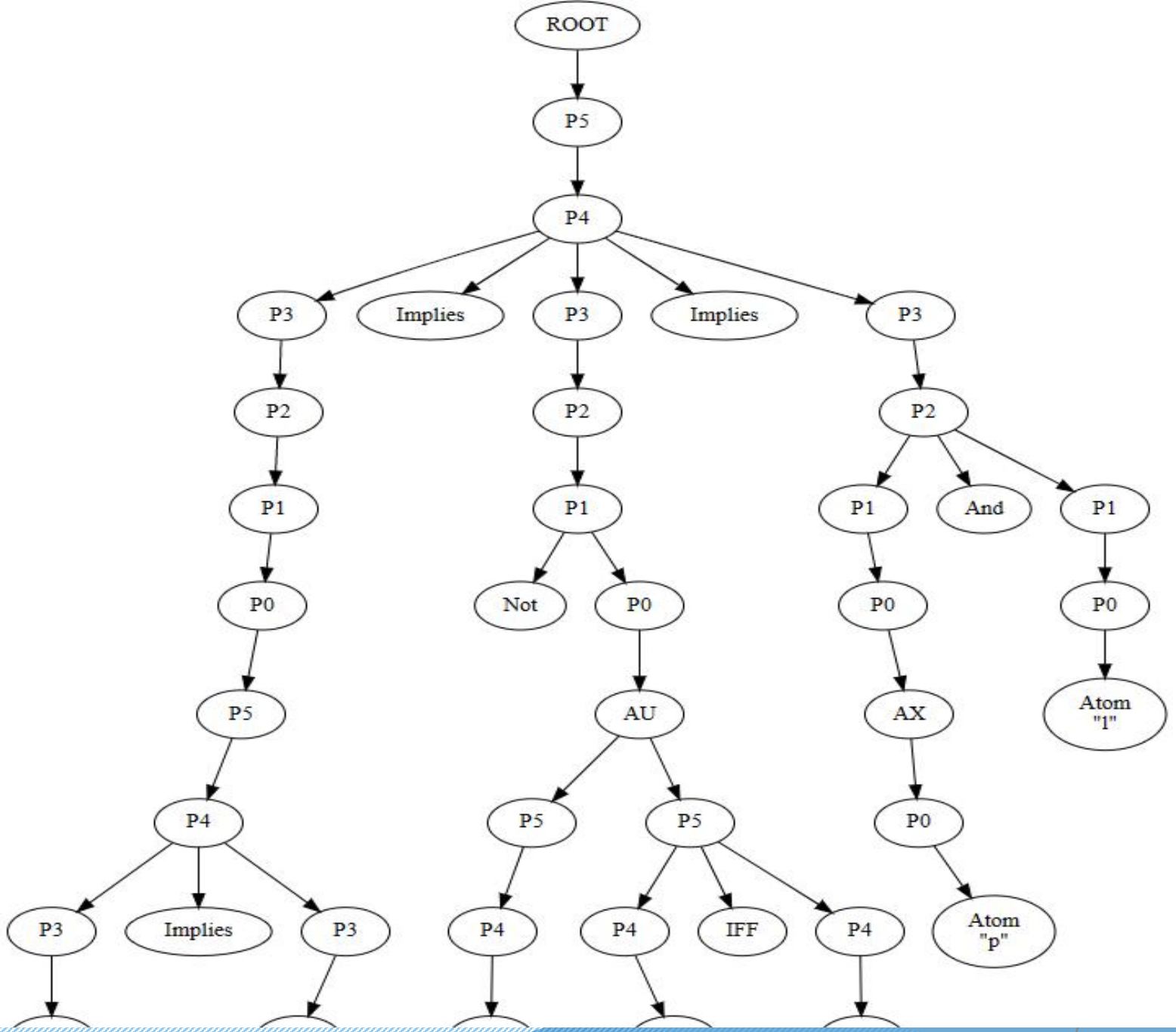
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What we did

2. Optimize AST

example:

((true⇒EX(¬f))

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⇒(AXp∧l)

To get a good AST, we need to do a lot of optimizations and transformations.

Here is the precessing:


parse_tree_
good_node


parse_tree_
no_text.svg


parse_tree_
full.svg

What we want to do next step?

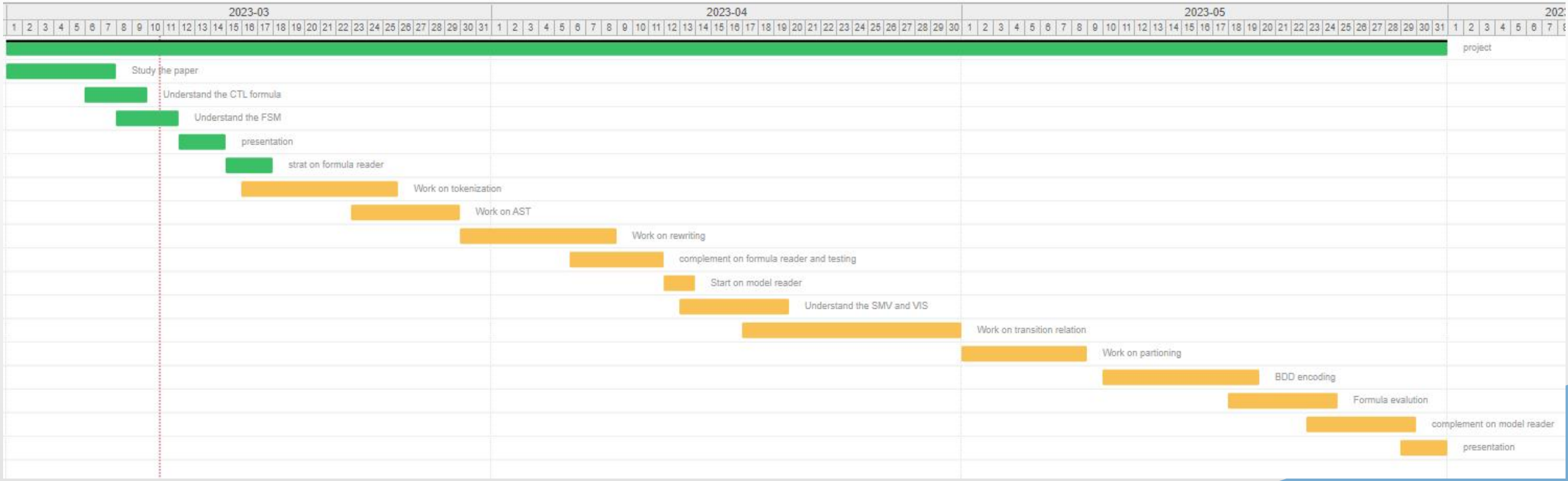
FIRST:

Try to finish the part of CTL formula reader: Tokenization, Optimize AST, rewriting

Second:

About the part of model reader, fous on partitioning and BDD encoding.

Schedule


2023-03
2023-04
2023-05
project
Study the paper
Understand the CTL formula
Understand the FSM
presentation
strat on formula reader
Work on tokenization
Work on AST
Work on rewriting
complement on formula reader and testing
Start on model reader
Understand the SMV and VIS
Work on transition relation
Work on partioning
BDD encoding
Formula evalution
complement on model reader
presentation

THANK YOU